Deep Convolutional Neural Network for Survival Analysis with Pathological Images

Xinliang Zhu, Jiawen Yao and Junzhou Huang
Department of Computer Science and Engineering
The University of Texas at Arlington, Arlington, TX, USA
Email: {xinliang.zhu, jiawen.yao}@mavs.uta.edu, jzhuang@uta.edu

***Abstract*—Traditional Cox proportional hazard model for survival analysis are based on structured features like patients' sex, smoke years, BMI, etc. With the development of medical imaging technology, more and more unstructured medical images are available for diagnosis, treatment and survival analysis. Traditional survival models utilize these unstructured images by extracting human-designed features from them. However, we argue that those hand-crafted features have limited abilities in representing highly abstract information. In this paper, we for the first time develop a deep convolutional neural network for survival analysis (DeepConvSurv) with pathological images. The deep layers in our model could represent more abstract information compared with hand-crafted features from the images. Hence, it will improve the survival prediction performance. From our extensive experiments on the National Lung Screening Trial (NLST) lung cancer data, we show that the proposed DeepConvSurv model improves significantly compared with four state-of-the-art methods.**

***Keywords*—Survival analysis; Deep learning; Lung cancer; Pathological images**

I. Introduction

Survival analysis in healthcare research aims at modeling the time that elapses from the beginning of follow-up until a certain event of interest (e.g. biological death) occurs. The prognostic models generated by survival prediction can be used to explore interaction between prognostic factors in certain diseases, and also predict how a new patient will behave in the context of known patients' healthcare data. The employment of survival prediction would allow clinicians to make early decisions on treatment which is very crucial for patients' healthcare.

In survival analysis, the most popular model is the Cox proportional hazards model [1]. The Cox model and its extensions have been extensively studied in the fields of statistical learning in recent decades [2], [3], [4], [5], [6]. However, these methods are built on the proportional hazards hypothesis which assumes that the hazard ratio between two instances is constant in time. This hypothesis further indicates that a patient's risk of death is a linear combination of covariates. However, there are too many complex interactions in real-world applications. The more comprehensive survival model is needed to better fit realistic data with nonlinear risk functions. To formulate the survival problem without any additional hypothesis, Li et al. modeled the prediction problem as the standard multi-task learning using an additional indicator matrix [7]. However, the number of task corresponds to the maximum follow-up time of all the instances and the multi-task learning will encounter computation issues if the follow-up time is large as it needs to learn a shared representation across all tasks at different time intervals.

Nowadays, deep learning has been a very successful technique in the field of machine learning and computer vision. It achieved best performances in many applications like image classification and recognition [8], [9], [10], [11], [12]. It has shown the capability to train a very complex model from practical dataset and extract very high-level features. Although deep learning has been applied in many applications, there's still very few work focusing on survival analysis. Katzman $et\ al.$ [13] was the first one proposing a deep fully connected network (DeepSurv) to model the nonlinear relations between the covariates and risk function. By replacing the linear part $\beta^T x$ in $f(x)$ with nonlinear deep fully connected network in DeepSurv, it showed its advantages in survival modeling from both simulated data and real survival data. However, DeepSurv is limited to process structured data like clinical information. For the unstructured data, which is not organized in a predefined manner, it can't handle them directly. Fig.1 shows the samples of structured data and unstructured data.

Recent work [14], [15], [16], [12] have introduced pathological images in lung cancer survival analysis and demonstrated that images can be used as additional information for survival analysis. However, they used hand-crafted image features to describe tumor morphology which are with limited representation abilities and their proposed methods are actually not end-to-end learning. Deep learning models, due to its hierarchical property, can represent much more abstract features compared with human-design ones. Thus we adopt it to survival analysis problem in our paper.

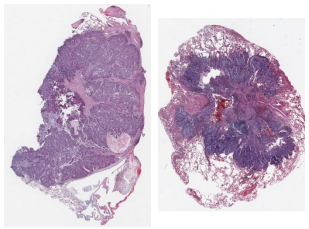
In this paper, we for the first time develop a DeepConvSurv model to do end-to-end survival analysis with unstructrued pathological images. It takes pathological images with variable size as inputs. With negative partial log likelihood as its loss function, it learns a deep convolutional neural network for survival analysis. The output of DeepConvSurv is the risk value of Cox model's exponential part. Thus the contributions of this paper are as follows: (1) we for the first time develop a DeepConvSurv model to predict patient's survival directly from the pathological images; (2) extensive experiments on NLST lung cancer dataset are conducted to evaluate the performance of proposed DeepConvSurv model. The experimental

978-1-5090-1610-5/16/$31.00 ©2016 IEEE

Authorized licensed use limited to: UNIVERSITAT OBERTA DE CATALUNYA. Downloaded on March 11,2026 at 13:00:53 UTC from IEEE Xplore. Restrictions apply.

smokeage	smokeday	cigsmok	gender	rndgroup	age	pkyr
22	20	1	2	1	61	37
13	60	0	1	1	68	132
14	40	0	1	1	65	94
21	40	0	2	1	60	58
16	25	1	2	1	60	55
15	30	1	1	2	68	79.5
13	40	0	1	1	61	88
19	30	1	1	1	59	60
16	40	0	1	1	63	74
17	20	1	1	1	60	43
19	40	1	1	1	59	80
18	20	0	1	1	57	35
14	40	1	1	1	62	96
15	20	0	1	1	72	52
15	30	1	2	2	59	66
20	40	1	2	2	60	80
14	20	1	2	1	73	59

Structured Data



Unstructured Data

Fig. 1. Structured data v.s Unstructured data: structured data in the left part is with a high level of organization, while the unstructured pathological images in the right part are with different sizes and not organized in a predefined manner.

results show that DeepConvSurv model outperform those state-of-the-art methods significantly, which demonstrates the powerful ability of the developed DeepConvSurv model.

II. Related Work

In this section, we give a brief survey on the classic survival modeling method, deep convolutional neural network, and the recently developed survival analysis method.

A. Traditional Survival Analysis

Survival analysis is about predicting the time duration until an event occurs, and in our case the event is the death of a lung cancer patient. Traditional survival modeling assumes the time durations follow an unknown distribution. In the modeling methods, Cox proportional hazard model is among the most popular one. It models the hazards instead of the survival function:

$$h(t|x) = h_0(t)exp(\beta^T x), \tag{1}$$

where t is the time, $x = (x_1, ...x_p)^T$ is the covariates of dimension p, $\beta = (\beta_1, ..., \beta_p)^T$ is a vector of regression parameters, and $h_0(t)$ is the baseline hazard. $f(x) = \beta^T x$ is also being called as risk function. The regression parameters can be estimated by minimizing negative log partial likelihood, which is

$$l(\beta) = -\sum_{i=1}^{n} \delta_i \left(\beta^T x_i - log \sum_{j \in R(t_i)} exp(\beta^T x_j) \right), \tag{2}$$

where n is the number of patients, t_i is the censored or observed survival time for patient i, and δ_i is an indicator for whether the survival time is censored ($\delta_i = 0$) or observed ($\delta_i = 1$). $R(t_i)$ is the risk set at time t_i, which is the set of all individuals who are still under study before time t_i.

However, traditional Cox model only could handle the linear condition in the risk function. If the covariates are nonlinear with the risk function, this model would not do the job well.

B. Deep Survival Analysis

Katzman *et al.* [13] proposed a deep fully connected network (DeepSurv) to represent the nonlinear relations between the covariates and risk function. It replaces the exponential part $\beta^T x$ in $f(x)$ with nonlinear deep fully connected network in traditional Cox model. From the experiments, they demonstrated that DeepSurv not only worked well in standard linear Cox proportional hazard model but also as well in the nonlinear settings. They conducted experiments on both simulated data and real survival data. However, this model is limited to process with structured data. It can't handle the unstructured images in the real applications well.

III. Proposed Methods

A. DeepConvSurv

1) The Model: Compared to hand-crafted features, deep learning models could represent the observed data as hierarchical abstraction at many different levels. In DeepConvSurv, we replace the exponential part of Cox model with deep convolutional network so that it could make survival prediction from pathological images directly.

Fig.2 shows the DeepConvSurv model. The input layer is a 1024×1024 patch extracted from the regions of interest(ROI) in pathological images. The ROI were annotated by human beings under the train of pathologists. It consists of 3 convolutional layer, 2 max-pooling layers and 1 fully-connected layer. In each convolutional layer, we employ ReLU as the nonlinear activation function.

2) Loss Function: Since we are conducting survival analysis based on Cox proportional hazard model, the main difference between our model with traditional Cox model relies on the exponential part. Thus the loss function of our model is:

$$l(w,b) = -\sum_{i:R_i=1} h_{w,b}^{last}(x_i) + log \sum_{j:t_j>=t_i} exp(h_{w,b}^{last}(x_j)), \tag{3}$$

where $h_{w,b}^{last}(x)$ is the output of DeepConvSurv (e.g. the last layer's output), w, b are the network's parameters, and x_i, x_j are the ith and j^{th} inputs. For simplicity, we will omit each layer's parameters and write $h_{w,b}^{last}(x)$ as $h^{last}(x)$.

3) Optimization: We apply stochastic gradient descent method (SGD) to solve DeepConvSurv model. Since the loss function has been changed, during training DeepConvSurv, the loss function's gradient with respect to the output node in back propagation algorithm is changed to:

$$\delta^{last}(x_i) = \frac{\partial l(w,b)}{\partial h^{last}(x_i)} = -1 + \frac{exp(h^{last}(x_i))}{\sum_{j:t_j>=t_i} exp(h^{last}(x_j))}, \tag{4}$$

where $\delta^{last}(x_i)$ is also being called as sensitivity of the output neuron. The sensitivities of other neurons in layer l are kept unchanged.

IV. Experiments

We conduct extensive experiments on NLST dataset to evaluate the proposed method. We compare DeepConvSurv with three state-of-the-art methods in survival analysis — Cox+SuperPC [3], Cox+Lasso [5] , random survival forests (RSF) [17], and DeepSurv [13]. SuperPC and Lasso are used for features dimension reduction. They are popular and achieving highest performance in traditional survival prediction. RSF

Authorized licensed use limited to: UNIVERSITAT OBERTA DE CATALUNYA. Downloaded on March 11,2026 at 13:00:53 UTC from IEEE Xplore. Restrictions apply.

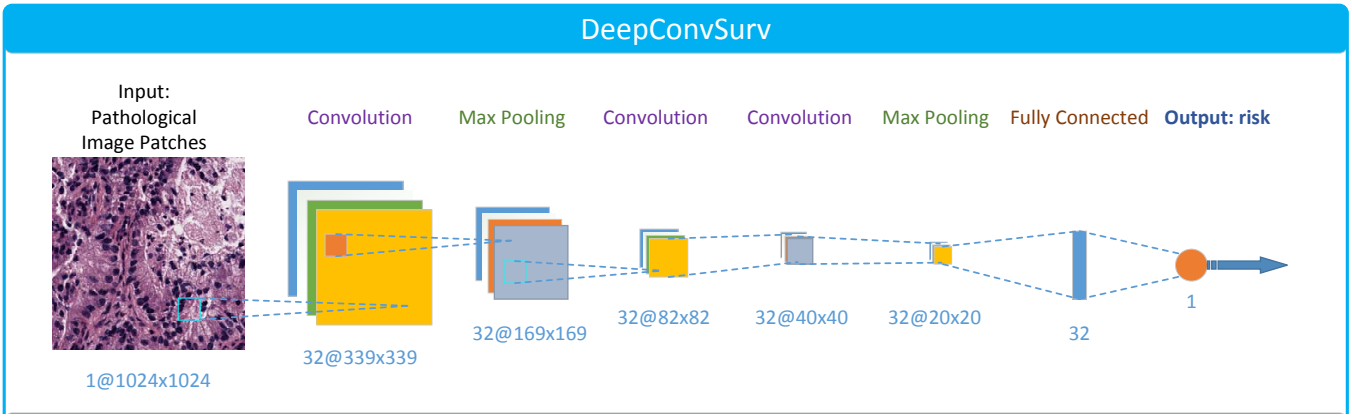


Fig. 2. DeepConvSurv model: DeepConvSurv model is based on deep convolutional neural network, however, the loss function of the network has been changed to negative partial log likelihood and the network has been set to work on the pathological image patches.

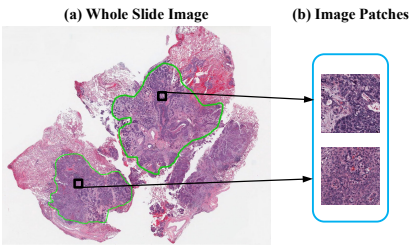


Fig. 3. Sample patches from a whole slide: This whole slide has two ROIs. Each ROI is surrounded by green curves. One 1024×1024 patch is randomly cut from each ROI.

boosts the survival prediction accuracy by employing random forest strategy. DeepSurv is a state-of-the-art deep model for analyzing structured survival data.

A. NLST Lung Cancer Dataset

In the NLST lung cancer dataset, 450 lung cancer patients with both pathological images and detailed clinical information were selected. The pathological images are with different size and they are too large to be processed by traditional methods directly.

B. Data Preprocessing

Pathological images from NLST dataset are whole slides. They are too big (with more than 100 million pixels per image) to be processed by a convolutional neural network. Thus we randomly sample some smaller size patches from the whole slides to represent them. The patches are extracted randomly from the region of interests (ROIs) annotated under the help of pathologists. Fig.3 shows the relations between whole slides, ROIs and the patches.

There're total 450 patients diagnosed with lung cancer having survival information. However, among them 46 are not dead from lung cancer directly, so we remove their data during the experiment. All the left 404 patients totally have 867 pathological images available. There're total 1031 ROIs from the 867 images. Finally, we extract one patch from each ROI and assign the survival information to the patches by patients ID.

To evaluate whether the proposed DeepConvSurv model is superior than hand-crafted features in extracting survival related information, we also use CellProfiler to generate human-design features motivated by recent work [18]. It is a free open-source software designed to enable researchers to quantitatively measure phenotypes from images automatically [19]. Due to its powerful and convenient modules in processing medical images, CellProfiler serves as a state-of-the-art feature extracting and quantitative analysis tool. The features are 1795 dimensions, which include texture features, geometry features and holistic statistics.

C. Experiment Setup

We employs 5-fold cross-validation to evaluate the proposed model and other state-of-the-art methods. The state-of-the-art methods used in our experiments are: Cox model with Lasso feature selection, Cox model with SuperPC feature transformation, RSF and DeepSurv. The training is very fast due to the simple network architecture. It takes less than 20 seconds to finish one training epoch using Nvidia Titan GPU. We fixed the total training epoch as 15 to achieve the best performance. So the training time is not an issue in our application.

To compare the performances in survival prediction, we take the standard performance measure for model assessment in survival analysis [20] concordance index (CI) as our evaluation metric. The CI is defined as follows:

$$CI = \frac{\#correctly\ ordered\ pairs}{\#all\ possible\ ranking\ pairs}. \tag{5}$$

Eq.(5) is the fraction of all pairs of subjects whose survival times are correctly ordered among all possible ranking pairs. It quantifies the ranking quality of rankings [20]. The value of 0 is with the worst performance, 0.5 is a random guess and 1 is the prefect condition. Cox proportional hazards model shows a relative risk of a patient compared with the other. So it is proper to adopt the CI as an evaluation index.

Authorized licensed use limited to: UNIVERSITAT OBERTA DE CATALUNYA. Downloaded on March 11,2026 at 13:00:53 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CI VALUES OF DIFFERENT MODELS IN PREDICTING SURVIVAL: HIGHER VALUE IS BETTER

Data	Model	CI
Images	Cox + LASSO [5]	0.562
	Cox + SuperPC [3]	0.556
	RSF [17]	0.511
	DeepSurv [13]	0.602
	DeepConvSurv	**0.629**

D. Results and Discussion

The results are shown in table I. It shows that deep learning based models are with higher CI values. As mentioned above, Cox model is about relative risk between two patients, so it's a ranking problem. The higher CI value means higher accuracy on ranking orders. Hence, the model with larger CI value has better performance in survival prediction.

From the experiments on predicting survival with images alone, DeepConvSurv performs much better than RSF and Cox model with Lasso and SuperPC. It's more than 12% improvement in CI value. It is also better than DeepSurv model using features extracted from CellProfiler. This shows that, from the perspective of features extracting, DeepConvSurv can catch more survival related features compared with the human-design method.

Thus, the main contributions of this paper are as follows:

- To improve the survival prediction performance with human-designed image features, we develop a DeepConvSurv model to do end-to-end survival prediction from pathological images.
- We conduct extensive experiments with NLST lung cancer dataset to assess the developed models' performance. The results show the developed model outperforming the state-of-the-art methods significantly.

V. CONCLUSION

In this paper, we develop a novel convolutional neural network — DeepConvSurv for survival prediction with pathological images. As far as we know, our work was the first one proposing deep convolutional neural networks for improving survival prediction performance with pathological images. The extensive experiments also showed the convolutional neural network's superiority in predicting survival. The proposed method paved a way using convolultional neural network to extract more predictive information in survival analysis. In the future, we expect to do more research on other datasets.

ACKNOWLEDGMENT

The authors would like to thank the National Cancer Institute for access to NCI's data collected by the National Lung Screening Trial. The statements contained herein are solely those of the authors and do not represent or imply concurrence or endorsement by NCI. We also thank Nvidia for supporting our research with NVIDIA Tesla K40 GPUs. This work was partially supported by U.S. NSF IIS-1423056, CMMI-1434401, CNS-1405985 and the NSF CAREER grant IIS-1553687.

REFERENCES

[1] D. R. Cox, "Regression models and life-tables," *Journal of the Royal Statistical Society. Series B (Methodological)*, pp. 187–220, 1972.

[2] E. Bair and R. Tibshirani, "Semi-supervised methods to predict patient survival from gene expression data," *PLoS Biol*, vol. 2, no. 4, p. E108, 2004.

[3] E. Bair, T. Hastie, D. Paul, and R. Tibshirani, "Prediction by supervised principal components," *Journal of the American Statistical Association*, vol. 101, no. 473, 2006.

[4] H. C. van Houwelingen, T. Bruinsma, A. A. Hart, L. J. van't Veer, and L. F. Wessels, "Cross-validated cox regression on microarray gene expression data," *Statistics in medicine*, vol. 25, no. 18, pp. 3201–3216, 2006.

[5] M. Y. Park and T. Hastie, "L1-regularization path algorithm for generalized linear models," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 69, no. 4, pp. 659–677, 2007.

[6] H. M. Bøvelstad, S. Nygård, H. L. Størvold, M. Aldrin, Ø. Borgan, A. Frigessi, and O. C. Lingjærde, "Predicting survival from microarray dataa comparative study," *Bioinformatics*, vol. 23, no. 16, pp. 2080–2087, 2007.

[7] Y. Li, J. Wang, J. Ye, and C. K. Reddy, "A multi-task learning formulation for survival analysis," in *In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD'16)*, 2016, pp. 1715–1724.

[8] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "Imagenet: A large-scale hierarchical image database," in *Computer Vision and Pattern Recognition, 2009. CVPR 2009. IEEE Conference on*. IEEE, 2009, pp. 248–255.

[9] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A. C. Berg, and L. Fei-Fei, "ImageNet Large Scale Visual Recognition Challenge," *International Journal of Computer Vision (IJCV)*, vol. 115, no. 3, pp. 211–252, 2015.

[10] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," *arXiv preprint arXiv:1512.03385*, 2015.

[11] K. H. J. S. Jifeng Dai, Yi Li, "R-FCN: Object detection via region-based fully convolutional networks," *arXiv preprint arXiv:1605.06409*, 2016.

[12] S. Wang, J. Yao, Z. Xu, and J. Huang, "Subtype cell detection with an accelerated deep convolution neural network," in *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. Springer, 2016, pp. 640–648.

[13] J. Katzman, U. Shaham, A. Cloninger, J. Bates, T. Jiang, and Y. Kluger, "Deep survival: A deep cox proportional hazards network," *arXiv preprint arXiv:1606.00931*, 2016.

[14] H. Wang, F. Xing, H. Su, A. Stromberg, and L. Yang, "Novel image markers for non-small cell lung cancer classification and survival prediction," *BMC Bioinformatics*, vol. 15, no. 1, p. 310, 2014. [Online]. Available: http://www.biomedcentral.com/1471-2105/15/310

[15] X. Zhu, J. Yao, X. Luo, G. Xiao, Y. Xie, A. Gazdar, and J. Huang, "Lung cancer survival prediction from pathological images and genetic data - an integration study," in *2016 IEEE 13th International Symposium on Biomedical Imaging (ISBI)*, 2016, pp. 1173–1176.

[16] J. Yao, S. Wang, X. Zhu, and J. Huang, "Imaging biomarker discovery for lung cancer survival prediction," in *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. Springer International Publishing, 2016, pp. 649–657.

[17] H. Ishwaran, U. B. Kogalur, E. H. Blackstone, and M. S. Lauer, "Random survival forests," *The annals of applied statistics*, pp. 841–860, 2008.

[18] J. Yao, D. Ganti, X. Luo, G. Xiao, Y. Xie, S. Yan, and J. Huang, "Computer-assisted diagnosis of lung cancer using quantitative topology features," in *Machine Learning in Medical Imaging*, ser. Lecture Notes in Computer Science, L. Zhou, L. Wang, Q. Wang, and Y. Shi, Eds. Springer International Publishing, 2015, vol. 9352, pp. 288–295.

[19] A. E. Carpenter, T. R. Jones, M. R. Lamprecht, C. Clarke, I. H. Kang, O. Friman, D. A. Guertin, J. H. Chang, R. A. Lindquist, J. Moffat *et al.*, "Cellprofiler: image analysis software for identifying and quantifying cell phenotypes," *Genome biology*, vol. 7, no. 10, p. R100, 2006.

[20] H. Steck, B. Krishnapuram, C. Dehing-oberije, P. Lambin, and V. C. Raykar, "On ranking in survival analysis: Bounds on the concordance index," in *Advances in neural information processing systems*, 2008, pp. 1209–1216.

Authorized licensed use limited to: UNIVERSITAT OBERTA DE CATALUNYA. Downloaded on March 11,2026 at 13:00:53 UTC from IEEE Xplore. Restrictions apply.